

OUTPUT

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IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Madhu Bavana/AppData/Local/Programs/Python/Python314/OUTPUT SCENARIO BASED.py
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SCENARIO 1: Flood-Relief Drone Routing
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Greedy Best-First -> nodes expanded: 29, path length: 29, true cost: 45.0
A* Search -> nodes expanded: 156, path length: 29, true cost: 28.0

--- With dynamic blockages appearing mid-search ---
Greedy (dynamic) -> nodes expanded: 29, path found: True
A* (dynamic) -> nodes expanded: 147, path found: True, cost: 28.0

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SCENARIO 2: Smart-City Traffic Signal Optimization
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Hill Climbing -> avg cost over 8 runs: -37.45, best: -37.96 (frequently stuck at local optima)
Simulated Annealing -> avg cost over 8 runs: -37.65, best: -37.96 (consistently escapes local optima)
SA improved average cost over Hill Climbing by ~-0.5%

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SCENARIO 3: Mars Rover Online Search (Partial Observability)
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Rover reached goal: True
Steps taken (online, no full map known in advance): 38
Total hazards in world: 54, hazards discovered via sensing: 13
Note: an OFFLINE search (e.g. plain A*) would require the full map upfront, which is unrealistic for real Mars exploration.

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SCENARIO 4: University Exam Timetabling (CSP)
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Backtracking + Forward Checking result:
Bio150 -> HallB, Mon-9am
CS101 -> HallA, Tue-9am
CS102 -> HallA, Tue-2pm
Chem110 -> HallB, Tue-9am
Math101 -> HallA, Mon-9am
Physics201 -> HallA, Mon-2pm
Nodes explored: 6

Plain Backtracking nodes explored: 21
Forward checking reduced search effort by 15 node visits.

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SCENARIO 5: Strategic Game AI (Minimax & Alpha-Beta Pruning)
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Plain Minimax -> best move: 7, score: 0, nodes explored: 205
Alpha-Beta Pruned -> best move: 7, score: 0, nodes explored: 123
Alpha-Beta pruning reduced node exploration by 40.0% while finding the SAME optimal move.

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